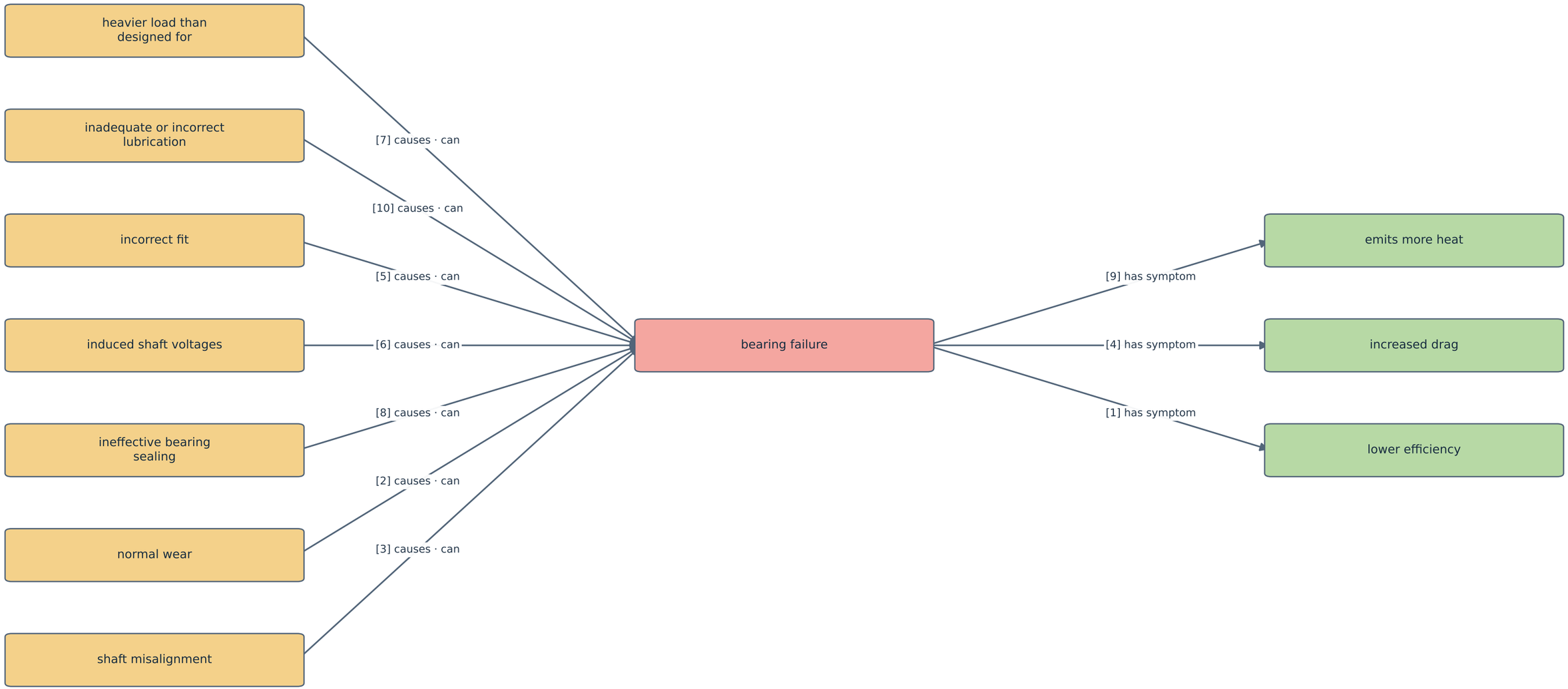
10 Bearing wear 13-common-causes.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode · Cause · Phenomenon · Detection Method · Maintenance Action
Component · Symptom

10 Bearing wear
13-common-causes.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] failed bearing → HAS_SYMPTOM → lower efficiency. Qualifier: asserted.
Page 6: has lower efficiency

[2] normal wear → CAUSES → bearing failure. Qualifier: can.
Page 6: Normal wear

[3] shaft misalignment → CAUSES → bearing failure. Qualifier: can.
Page 6: Shaft misalignment

[4] failed bearing → HAS_SYMPTOM → increased drag. Qualifier: asserted.
Page 6: A failed bearing has increased drag

[5] incorrect fit → CAUSES → bearing failure. Qualifier: can.
Page 6: Incorrect fit

[6] induced shaft voltages → CAUSES → bearing failure. Qualifier: can.
Page 6: Induced shaft voltages

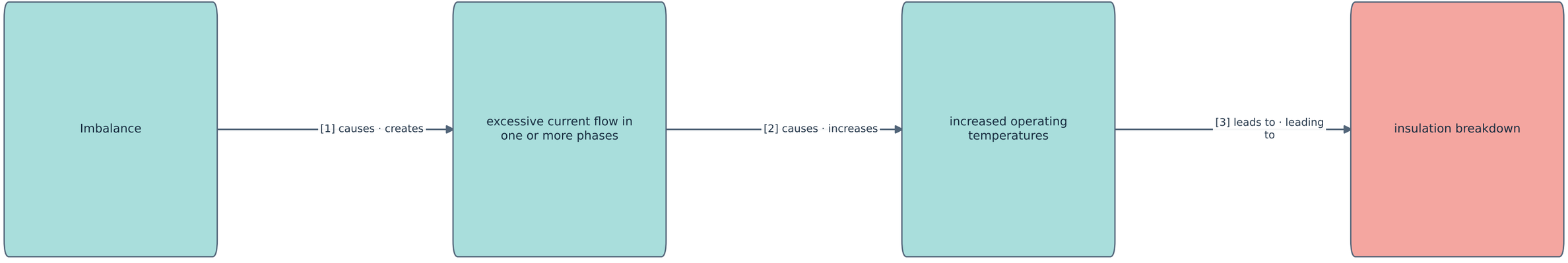
[7] heavier load than designed for → CAUSES → bearing failure. Qualifier: can.
Page 6: A heavier load than designed for

[8] ineffective bearing sealing → CAUSES → bearing failure. Qualifier: can.
Page 6: Ineffective bearing sealing

[9] failed bearing → HAS_SYMPTOM → emits more heat. Qualifier: asserted.
Page 6: emits more heat

[10] inadequate or incorrect lubrication → CAUSES → bearing failure. Qualifier: can.
Page 6: Inadequate or incorrect lubrication

2 Voltage imbalance 13-common-causes.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

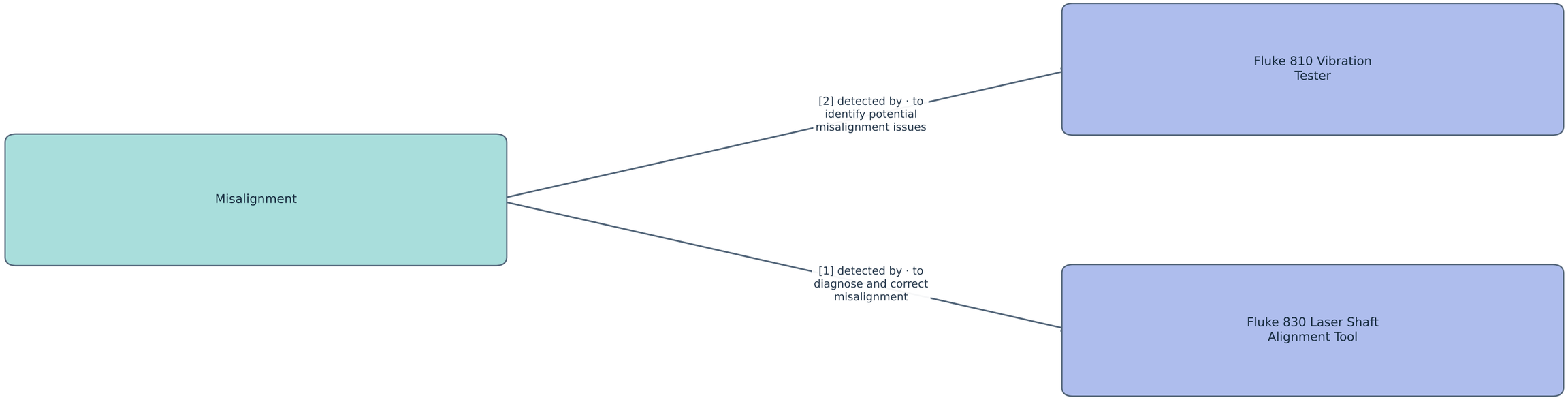
2 Voltage imbalance
13-common-causes.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] Imbalance → CAUSES → excessive current flow in one or more phases. Qualifier: creates.
Page 2: Imbalance creates excessive current flow in one or more phases

[2] excessive current flow in one or more phases → CAUSES → increased operating temperatures. Qualifier: increases.
Page 2: that then increases operating temperatures

[3] increased operating temperatures → LEADS_TO → insulation breakdown. Qualifier: leading to.
Page 2: leading to insulation breakdown

7 Misalignment 13-common-causes.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

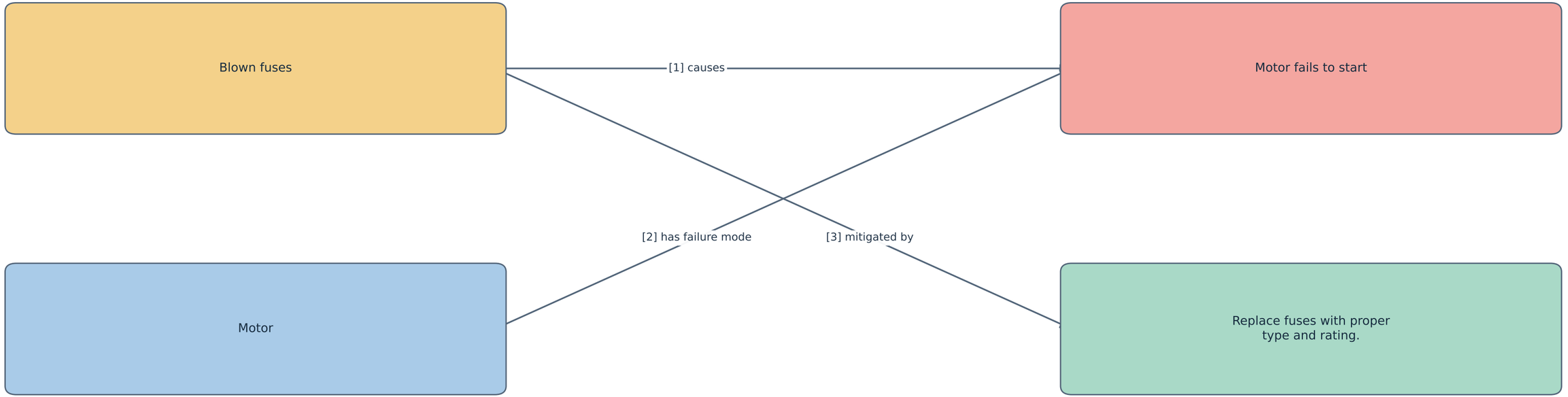
Failure Mode · Cause · Phenomenon · Detection Method · Maintenance Action
Component · Symptom

7 Misalignment
13-common-causes.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] Misalignment → DETECTED_BY → Fluke 830 Laser Shaft Alignment Tool. Qualifier: to diagnose and correct misalignment.
Page 5: Fluke 830 Laser Shaft Alignment Tool to diagnose and correct misalignment

[2] Misalignment → DETECTED_BY → Fluke 810 Vibration Tester. Qualifier: to identify potential misalignment issues.
Page 5: Fluke 810 Vibration Tester to identify potential misalignment issues

Motor fails to start — fuses 3GZF500730-85 RevH_EN 06-2022_lowres.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

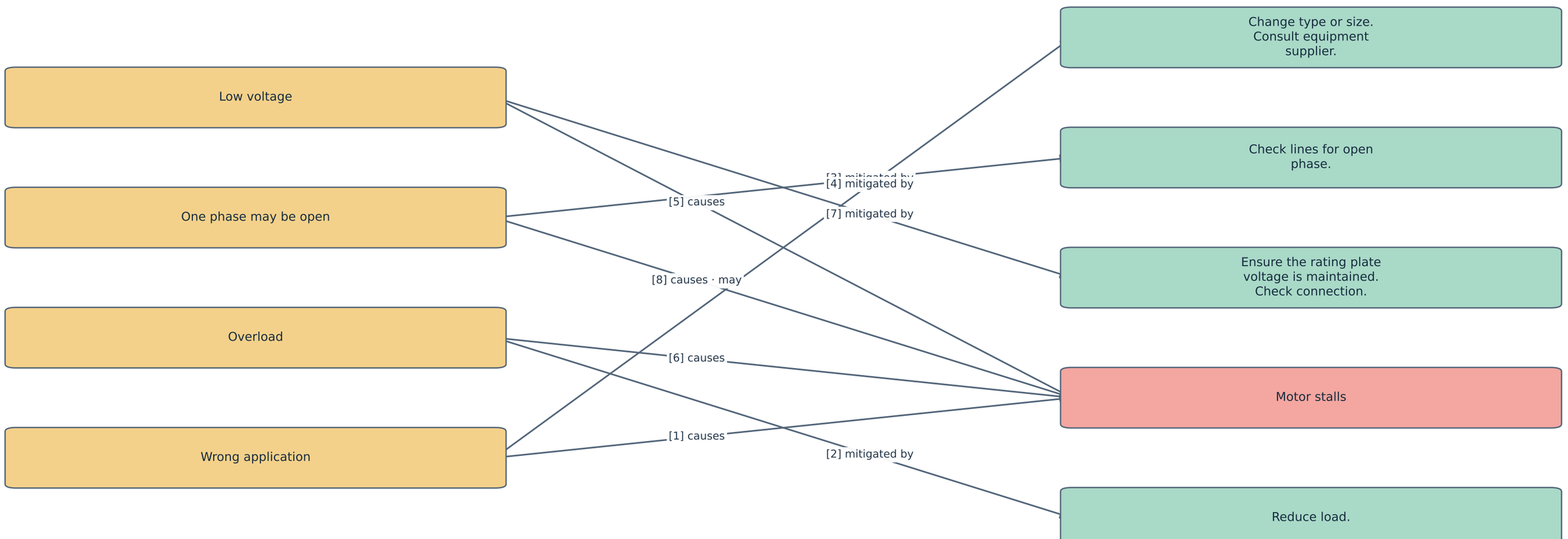
Motor fails to start — fuses
3GZF500730-85 RevH_EN 06-2022_lowres.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] Blown fuses → CAUSES → Motor fails to start. Qualifier: asserted.
Page 25: Blown fuses
Page 25: Motor fails to start

[2] Motor → HAS_FAILURE_MODE → Motor fails to start. Qualifier: asserted.
Page 25: Motor fails to start

[3] Blown fuses → MITIGATED_BY → Replace fuses with proper type and rating.. Qualifier: asserted.
Page 25: Replace fuses with proper type and rating.

Motor stalls 3GZF500730-85 RevH_EN 06-2022_lowres.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

Motor stalls
3GZF500730-85 RevH_EN 06-2022_lowres.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] Wrong application → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Wrong application
Page 25: Motor stalls

[2] Overload → MITIGATED_BY → Reduce load.. Qualifier: asserted.
Page 25: Reduce load.

[3] One phase may be open → MITIGATED_BY → Check lines for open phase.. Qualifier: asserted.
Page 25: Check lines for open phase.

[4] Wrong application → MITIGATED_BY → Change type or size. Consult equipment supplier.. Qualifier: asserted.
Page 25: Change type or size. Consult equipment supplier.

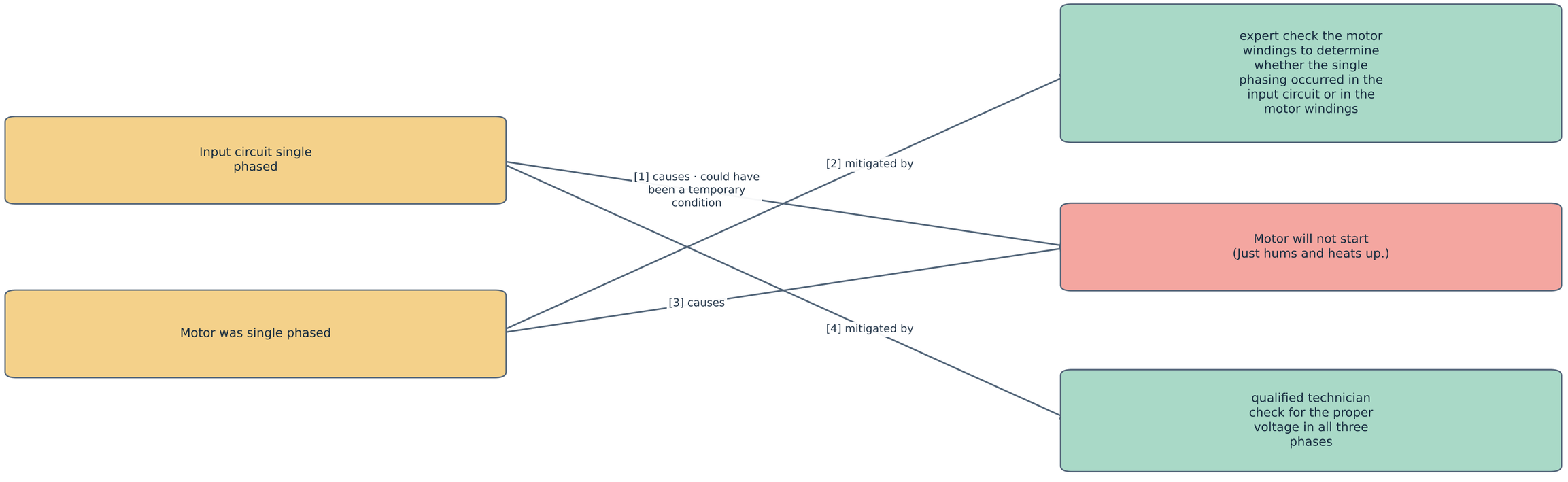
[5] Low voltage → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Low voltage
Page 25: Motor stalls

[6] Overload → CAUSES → Motor stalls. Qualifier: asserted.
Page 25: Overload
Page 25: Motor stalls

[7] Low voltage → MITIGATED_BY → Ensure the rating plate voltage is maintained. Check connection.. Qualifier: asserted.
Page 25: Ensure the rating plate voltage is maintained. Check connection.

[8] One phase may be open → CAUSES → Motor stalls. Qualifier: may.
Page 25: One phase may be open
Page 25: Motor stalls

Motor will not start; hums and heats Electric Motor Troubleshooting (Polyphase).pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

Motor will not start; hums and heats
Electric Motor Troubleshooting (Polyphase).pdf
Arrow references — qualifiers and conditions belong to the claim.

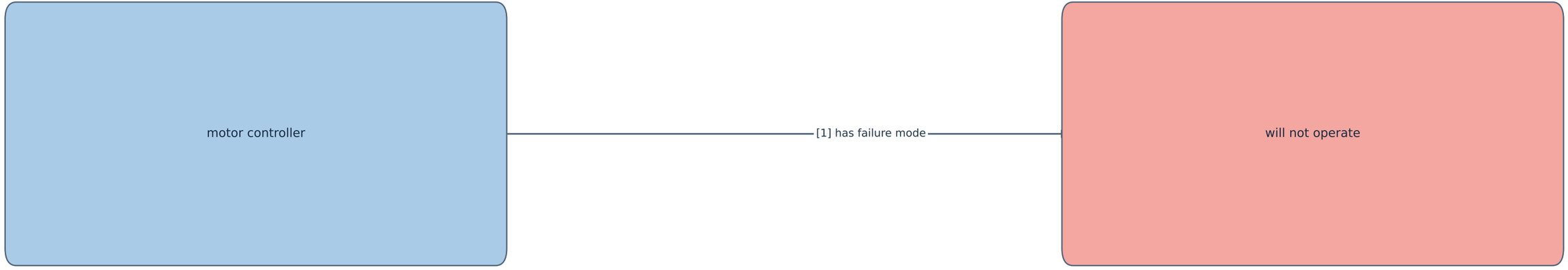
[1] Input circuit single phased → CAUSES → Motor will not start (Just hums and heats up.). Qualifier: could have been a temporary condition.
Page 6: Cause 3a: Input circuit single phased (No voltage in one or two phases; could have been a temporary condition).

[2] Motor was single phased → MITIGATED_BY → expert check the motor windings to determine whether the single phasing occurred in the input circuit or in the motor windings. Qualifier: asserted.
Page 6: Cure 3b: Replace the motor. ALWAYS have an expert check the motor windings to determine whether the single phasing occurred in the input circuit or in the motor windings. If in the input circuit, correct the single phase condition.

[3] Motor was single phased → CAUSES → Motor will not start (Just hums and heats up.). Qualifier: asserted.
Page 6: Cause 3b: Motor was single phased as indicated by heat damage to one or two phases of the motor windings.

[4] Input circuit single phased → MITIGATED_BY → qualified technician check for the proper voltage in all three phases. Qualifier: asserted.
Page 6: Cure 3a: Have a qualified technician check for the proper voltage in all three phases. If a single phase condition exists, correct the problem.

Starting: motor controller Electric Motor Troubleshooting (Polyphase).pdf



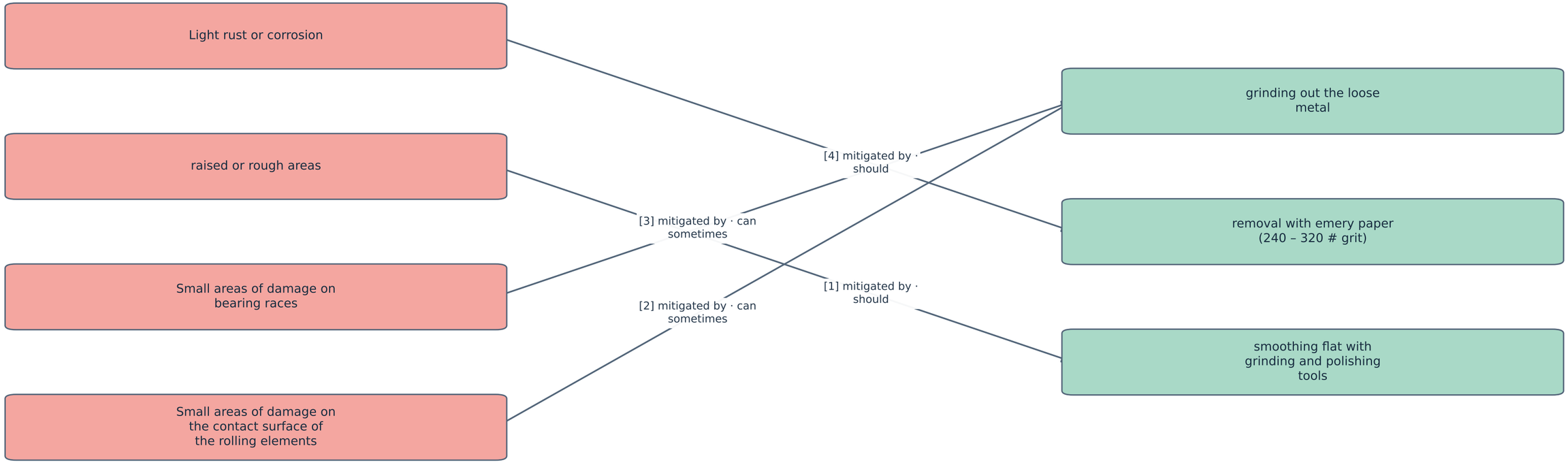
Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Component
Cause
Symptom
Phenomenon
Detection Method
Maintenance Action

Starting: motor controller
Electric Motor Troubleshooting (Polyphase).pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] motor controller → HAS_FAILURE_MODE → will not operate. Qualifier: asserted.
Page 6: Cause 2a: The motor controller will not operate.

Bearing repair Manual_Udrzba_Montaz_TIMKEN_AJ.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode · Cause · Phenomenon · Detection Method · Maintenance Action · Component · Symptom

Bearing repair
Manual_Udrzba_Montaz_TIMKEN_AJ.pdf
Arrow references — qualifiers and conditions belong to the claim.

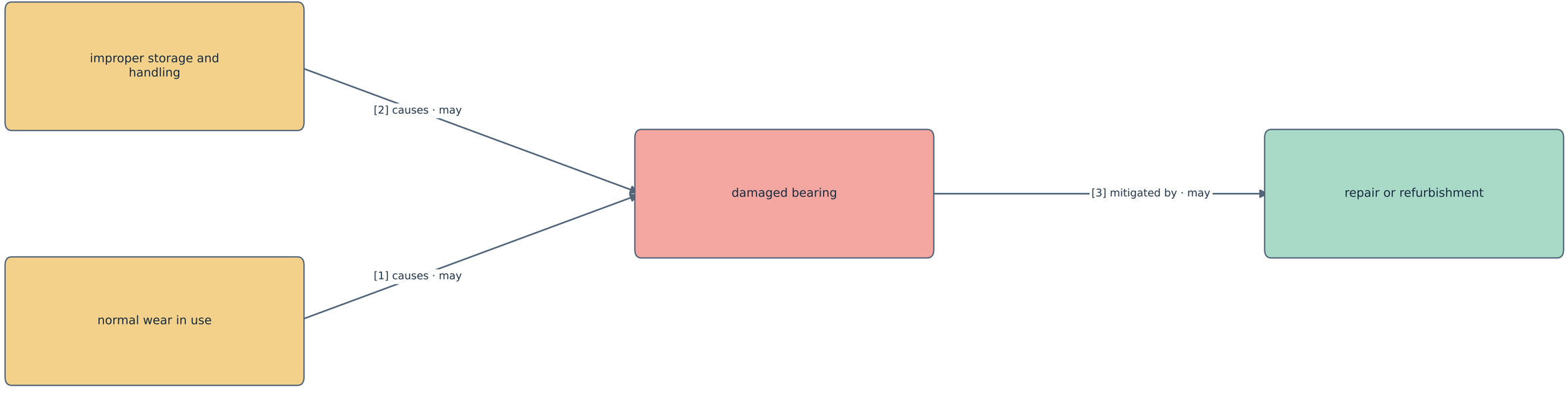
[1] raised or rough areas → MITIGATED_BY → smoothing flat with grinding and polishing tools. Qualifier: should.
Page 12: Any raised or rough areas should be smoothed flat with grinding and polishing tools.

[2] Small areas of damage on the contact surface of the rolling elements → MITIGATED_BY → grinding out the loose metal. Qualifier: can sometimes.
Page 12: Small areas of damage on bearing races, and on the contact sur- face of the rolling elements, can sometimes be repaired by grind- ing out the loose metal.

[3] Small areas of damage on bearing races → MITIGATED_BY → grinding out the loose metal. Qualifier: can sometimes.
Page 12: Small areas of damage on bearing races, and on the contact sur- face of the rolling elements, can sometimes be repaired by grind- ing out the loose metal.

[4] Light rust or corrosion → MITIGATED_BY → removal with emery paper (240 – 320 # grit). Qualifier: should.
Page 12: Light rust or corrosion should be removed with emery paper (240 – 320 # grit).

Damaged bearings Manual_Udrzba_Montaz_TIMKEN_AJ.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

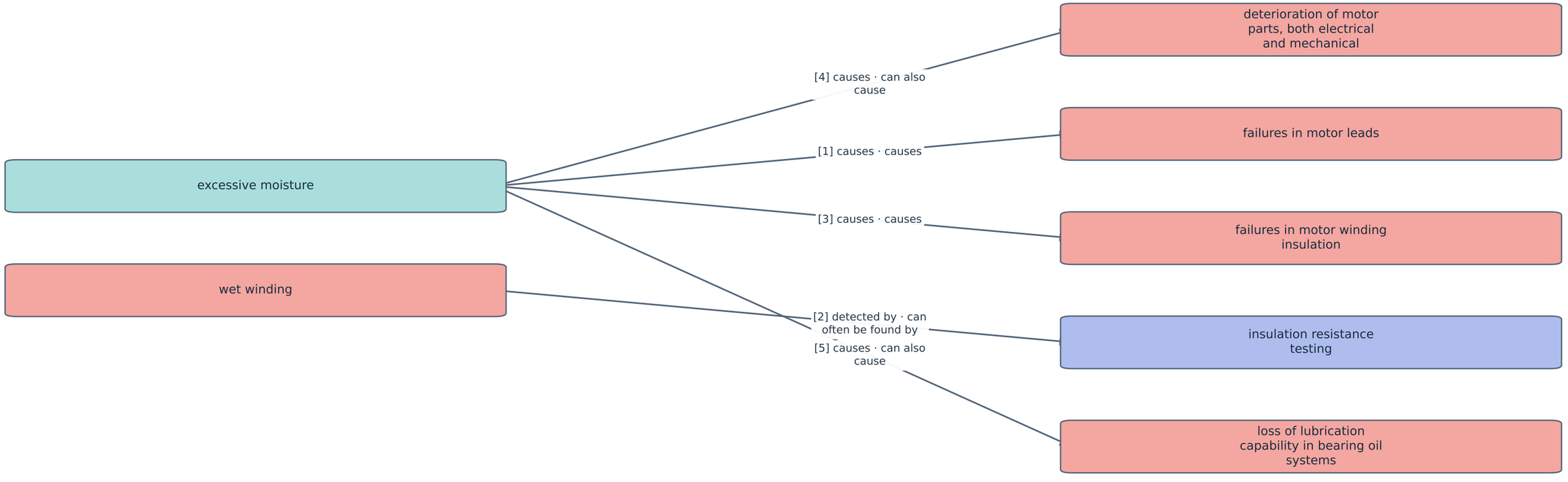
Damaged bearings
Manual_Udrzba_Montaz_TIMKEN_AJ.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] normal wear in use → CAUSES → damaged bearing. Qualifier: may.
Page 12: normal wear in use

[2] improper storage and handling → CAUSES → damaged bearing. Qualifier: may.
Page 12: improper storage and handling

[3] damaged bearing → MITIGATED_BY → repair or refurbishment. Qualifier: may.
Page 12: may still be returned to service after repair or refurbishment

Moisture and motor insulation NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode · Cause · Phenomenon · Detection Method · Maintenance Action · Component · Symptom

Moisture and motor insulation
NP_7502_Electric Motor Predictive and Preventive Maintenance Guide.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] excessive moisture → CAUSES → failures in motor leads. Qualifier: causes.
Page 54: Excessive moisture causes failures in motor winding insulation and in motor leads

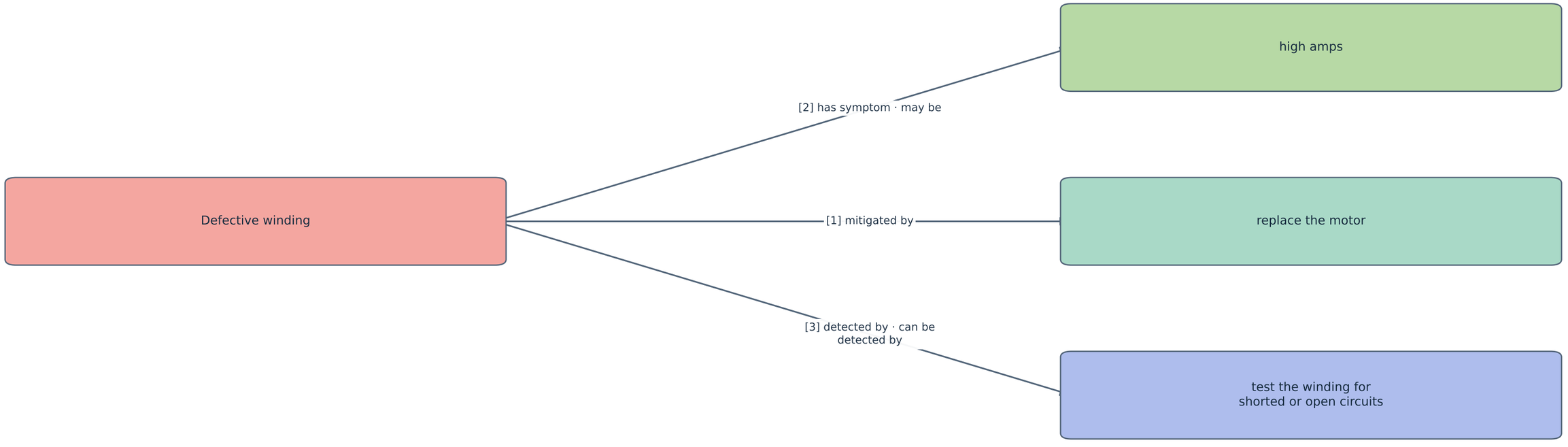
[2] wet winding → DETECTED_BY → insulation resistance testing. Qualifier: can often be found by.
Page 54: Indication of a wet winding can often be found by insulation resistance testing

[3] excessive moisture → CAUSES → failures in motor winding insulation. Qualifier: causes.
Page 54: Excessive moisture causes failures in motor winding insulation

[4] excessive moisture → CAUSES → deterioration of motor parts, both electrical and mechanical. Qualifier: can also cause.
Page 54: and deterioration of motor parts, both electrical and mechanical

[5] excessive moisture → CAUSES → loss of lubrication capability in bearing oil systems. Qualifier: can also cause.
Page 54: It can also cause loss of lubrication capability in bearing oil systems

Defective winding ac-troubleshooting-guide.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

Defective winding
ac-troubleshooting-guide.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] Defective winding → MITIGATED_BY → replace the motor. Qualifier: asserted.
Page 3: For defective winding, replace the motor.

[2] Defective winding → HAS_SYMPTOM → high amps. Qualifier: may be.
Page 3: The amps may also be high.

[3] Defective winding → DETECTED_BY → test the winding for shorted or open circuits. Qualifier: can be detected by.
Page 3: Test the winding for shorted or open circuits.

Unbalanced load and bearing failure ac-troubleshooting-guide.pdf



Draft source-supported claims · arrows retain source qualifiers · numbered evidence follows in PDF

Failure Mode
Cause
Phenomenon
Detection Method
Maintenance Action
Component
Symptom

Unbalanced load and bearing failure
ac-troubleshooting-guide.pdf
Arrow references — qualifiers and conditions belong to the claim.

[1] unbalanced load → CAUSES → bearings to fail. Qualifier: will also.
Page 3: An unbalanced load will also cause the bearings to fail.